

- 5 (a) Three resistors of resistances R_1 , R_2 and R_3 are connected in a network in parallel as shown in Figure 5.1.

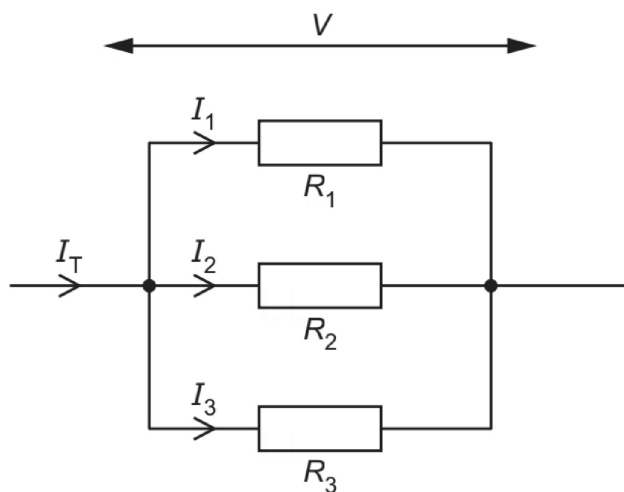


Figure 5.1

The currents in the resistors are I_1 , I_2 and I_3 respectively. The total current entering the network is I_T . The potential difference (p.d.) across the resistors is V .

Derive the equation

$$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

where R_T is the total resistance of the network.

- (b) Two identical resistance wires, each of resistance $2.8\ \Omega$, are connected in parallel to a battery with negligible internal resistance and electromotive force (e.m.f.) 12 V as shown in Figure 5.2.

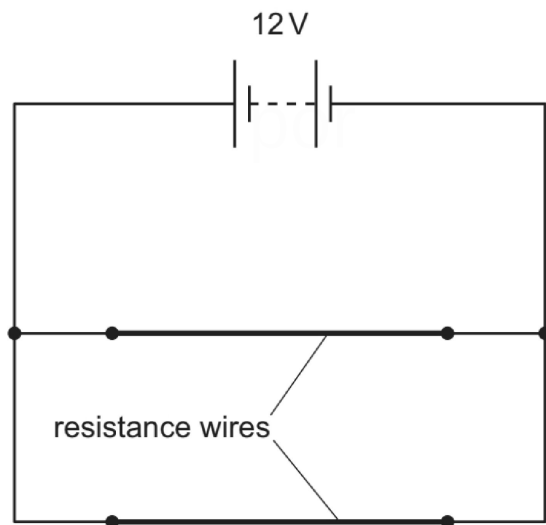


Figure 5.2

- (i) Calculate the current in the battery.

current =A [2]

- (ii) Each resistance wire has a cross-sectional area of $6.1 \times 10^{-6}\text{ m}^2$.

The wires are made from a metal with a number density of charge carriers (free electrons) of $8.5 \times 10^{28}\text{ m}^{-3}$.

Calculate the average drift speed of the charge carriers in the wires.

speed = m s^{-1} [2]

(c) A component X is added to the circuit in **5(b)** as shown in Figure 5.3.

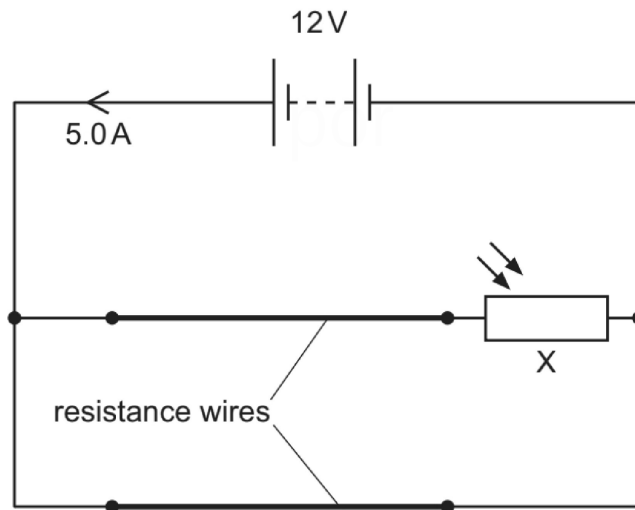


Figure 5.3

The current in the battery is now 5.0A.

The resistance of each resistance wire remains $2.8\,\Omega$.

(i) State the name of component X.

..... [1]

(ii) Determine the resistance of component X.

resistance = Ω [3]



- (iii) A change in the environment of component X causes the current in the battery to decrease. The resistance of the resistance wires remains constant.

State and explain how the environment of X has changed.

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..... [2]

[Total: 12]